

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of B.Sc. Examination, May, 2018

Course code & Name BS 401 & Foundations of Immunology

Date: 02/05/2018 Day: Wednesday Time: 10.00 AM to 10.20 AM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	Lysozyme:		01
	(i) Is a cytoplasmic organelle.	(ii) Activates complement.	
	(iii) Is a proteolytic enzyme.	(iv) Splits peptidoglycan.	
2.	The ability of the immune system to recognize self antigens versus nonself antigen is an example of:		01
	(i) Tolerance	(ii) Specific immunity	
	(iii) Cell mediated immunity	(iv) Humoral immunity	
3.	Which of the following is a non-organ-specific (systemic) autoimmune disease:		01
	(i) Myasthenia Gravis.	(ii) Insulin-dependent diabetes mellitus.	
	(iii) Pernicious anemia.	(iv) Systemic lupus erythematosus (SLE).	
4.	Monoclonal antibodies recognize a single:		01
	(i) Antigen	(ii) Epitope	
	(iii) B cell	(iv) Bacterium	

5.	The O blood group is the universal donor because the blood has		01
	(i) Antigen A	(ii) Antigen B	
	(iii) Both antigens A and B	(iv) No antigens	
6.	A living microbe with reduced virulence that is used for vaccination is considered:		01
	(i) A toxoid	(ii) Virulent	
	(iii) Dormant	(iv) Attenuated	
7.	Which of the following substances will not stimulate an immune response unless they are bound to a larger molecule?		01
	(i) Antigen	(ii) Hapten	
	(iii) Virus	(iv) Antibody	
8.	Tissue damage caused by wounds or invading pathogenic organisms induces a complex sequence of events collectively known as		01
	(i) Opsonization	(ii) Inflammation	
	(iii) Phagocytosis	(iv) None of the above	
9.	Cytokines are:		01
	(i) Hormone like polypeptides	(ii) Viruses	
	(iii) Bacteria	(iv) Carcinogens	
10.	B cells mature in the..... while T cells mature in the		01
	(i) Thymus/bone marrow and gut associated lymphoid tissue (GALT)	(ii) Bone marrow and GALT/Thymus	
	(iii) Spleen/bone marrow and GALT	(iv) Liver/Kidneys	

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IV Semester of B.Sc. Examination, May, 2018

Course code & Name BS 401 & Foundations of Immunology

Date: 02/05/2018 Day: Wednesday Time: 10.21 AM to 12.00 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed (attempt any 5)		10
1.	What are the complement proteins? What are the role of complement proteins in the immune system?	02
2.	What are the difference between passive and active immunity?	02
3.	How do vaccines give immunity?	02
4.	What are the difference Between MHC Class I and II?	
5.	How do you make monoclonal antibodies? How can monoclonal antibodies be used?	02
6.	Write a short note on Immunosuppression.	02
7.	What are the key cells of the immune system?	02
SECTION – II		
Q-III Answer the following questions as directed (attempt any 5)		15
1.	What is phagocytosis? Describe briefly the process.	03
2.	Construct a table comparing the biological characteristics of the five classes of immunoglobulins.	03
3.	What are cancers of the immune system? What kind of cancer can be treated with immunotherapy?	03
4.	What are the different types of hypersensitivity? Discuss briefly.	03

5.	What is Autoimmunity? What causes an autoimmune disease?	03
6.	What is transplantation? Why transplants are rejected?	03
7.	Write down the names of different immunodiagnostic techniques? Discuss briefly any one of them.	03
8.	What are the different phases of adaptive immune responses? Discuss briefly.	03